

EUCCLIDEAN ALGORITHM



Euclidean Algorithm

Suppose $a, b \in \mathbb{N}$, if we repeatedly perform the division algorithm:

$$a = bq_1 + r_1$$

$$b = r_1q_2 + r_2$$

$$r_1 = r_2q_3 + r_3$$

$\vdots$

$$r_{n-3} = r_{n-2}q_{n-1} + r_{n-1}$$

$$r_{n-2} = r_{n-1}q_n + 0 \quad (*)$$

then $r_{n-1} = \gcd(a, b)$.

proof: Notice that the sequence $\{r_1, r_2, \dots\} \subseteq \mathbb{N}$ and is decreasing. Thus it truncates, say at r_{n-1} . i.e. r_{n-1} exists.

Next: Show $r_{n-1} \mid a$ and $r_{n-1} \mid b$.

$(*) \Rightarrow r_{n-1} \mid r_{n-2} \Rightarrow r_{n-1} \mid r_{n-3} \Rightarrow \dots r_{n-1} \mid a$ and b .

Suppose $d \mid a$ and $d \mid b$. We want to show $d \mid r_{n-1}$, thus $r_{n-1} = \gcd(a, b)$.

Observe: $d|a$ and $d|b \Rightarrow d|a - bq_1 \Rightarrow d|r_1$

$$\Rightarrow d|b - r_1q_1$$

$$\Rightarrow d|r_2$$

$$\Rightarrow \dots \quad d|r_{n-1}$$

$$\Rightarrow r_{n-1} = \gcd(a, b)$$

Show $d|a, d|b \Rightarrow d|a - bq_1$.

$$d|a \text{ and } d|b \Rightarrow \exists k, l \in \mathbb{Z} \text{ with } dk = a, dl = b$$

$$\Rightarrow a - bq_1 = dk - dlq_1 = d(k - lq_1)$$

$$\Rightarrow d|a - bq_1$$